

Master Thesis — Progress Report

Spain's MTU15 reform: empirical assessment in progress

Pablo Paramio Pérez
Advisor: Pedro Mira

CEMFI

May 2026

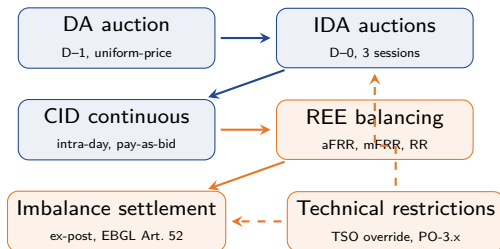
Outline

- Motivation
- Market structure and the 2024–2025 reform sequence
- Economic mechanisms in the literature
- Identification design and the data we use
- Main results and what they mean
- What is robust, what is not, and what is left

Motivation

- Empirically assess the impact of Spain's 2024–2025 **market time unit (MTU15) reform sequence**, focusing on *firm strategic behaviour* and *market outcomes*.
- Why this reform is interesting: it is a rare **natural experiment in market design**. It changes the time-unit granularity of the wholesale market in three stages, with *transitional asymmetric-clock windows* where settlement and trading run on different clocks.
- Why it is hard: the reform sequence is not cleanly separated from confounders — gas-price normalisation (Dec 2023), blackout (Apr 2025) and post-blackout *operación reforzada*, the elimination of CNMC Rule 28.8 (Mar 2025). **Institutional awareness is essential to read the data.**
- This talk: identification strategy I have developed, headline result on Big-4 strategic intraday repositioning, and an honest account of what survives robustness and what does not.

The Spanish wholesale market



- **OMIE (market operator)**: Day-ahead, intraday auctions, continuous intraday.
- **REE (TSO)**: balancing markets (reserves), imbalance settlement, technical restrictions.
- **Sequence**: forward markets get progressively closer to delivery; final adjustments via balancing.
- TSO can *override* OMIE programmes when network or stability constraints bind ⇒ especially relevant under *operación reforzada*.
- Firms have multiple instruments to take strategic positions across stages.

The 2024–2025 reform sequence

- Traditionally MTU = **60 min** everywhere.
- Three reform stages:
 - **Dec 2024** — **ISP15**: imbalance settlement → 15 min.
 - **Mar 2025** — **ID15**: intraday → 15 min.
 - **Oct 2025** — **DA15**: day-ahead → 15 min.
- Two transitional *asymmetric-clock* windows: settlement runs at finer granularity than trading.
- Plus: **2025-04-28 blackout** → *operación reforzada* (REE reinforced-operation regime), continuous from Apr 28 onward.

Period	ISP	ID	DA
Pre-Dec 2024	60	60	60
Dec24–Mar25	15	60	60
Mar25–Sep25	15	15	60
Oct25 onward	15	15	15

ISP15

Dec 2024

ID15

Mar 2025

DA15

Oct 2025

What does theory predict for finer time-unit granularity?

Three mechanisms drive expectations about the reform's effect:

Allaz & Vila (1993) — forward markets dispel rents

Adding forward markets prior to the spot stage commits firms to quantities and erodes their ability to exercise unilateral market power. *More finely-sliced spot markets imply more committable instruments* — so MTU15 should make withholding harder.

Ito & Reguant (2016, AER) — sequential markets and arbitrage

With market power, firms strategically reposition between sequential markets: under-commit in DA, sell more in IDA at higher prices. *Finer granularity in the second stage gives more strategic optionality within each hour.* The DiD prediction: critical-hour repositioning increases under symmetric-15-min clocks.

ISP15 mechanism — penalty for volatile technologies

Pre-ISP15: an hour's intra-hour deviations could net out (a 200 MW shortfall in Q1 cancels a 200 MW excess in Q3). Post-ISP15 they cannot. *Volatile technologies — wind, solar, fast-ramping CCGT — face strictly higher imbalance exposure.* Predicts behavioural response in critical hours where within-hour profile is steepest.

What we want to test, and how we test it

The outcomes we care about: firm strategic behaviour at the granularity-bite stages.

- q_1 = Big-4 day-ahead auction-cleared sell, MWh per firm-hour.
- q_2 = Big-4 *voluntary* IDA repositioning (signed PIBCIE), MWh per firm-hour. **Headline outcome.**
- Bid-shape: number of price tranches, ladder slope, qty-weighted price (post-MTU15-DA only).

Why MTU15 should affect critical hours more than flat hours: granularity reform only bites economically in hours where the within-hour residual-demand profile is *steep* — so 15-min slicing creates strategic optionality. In hours where the profile is *flat*, 15-min and 60-min are mechanically equivalent.

How we define them:

- **Treated** = critical hours $h \in \{7, 8, 16, 17, 18\}$. Top-5 by within-hour residual-demand variance σ_{within}^2 (morning ramp + evening peak).
- **Control** = flat hours $h \in \{3, 4, 5\}$. Bottom of σ_{within}^2 ranking (pre-dawn).

Why flat hours as counterfactual: they share the same day — so daily shocks (weather, demand level, fuel prices, regulatory announcements) absorb mechanically. If granularity reform has zero economic content in flat hours (because nothing within the hour to slice), then the within-day differential identifies the reform's effect on the granularity mechanism specifically.

The DiD specification

Specification:

$$q_{ift} = \alpha + \beta_1 \text{crit}_h + \beta_2 \text{post}_t + \beta_3 (\text{crit}_h \times \text{post}_t) + \gamma_f + \delta_d + \varepsilon_{ift}$$

β_3 is the DiD coefficient. Big-4 firm f , day t (Day-of-Week FE δ_d), hour h . Cluster-robust SE by date.

Window in each regression:

- **Pre** = pre-IDA: 2018-01-01 → 2024-06-13 (the structural baseline).
- **Post** = DA15/ID15: 2025-10-01 → present (the symmetric-clock regime).
- Intermediate regimes (3-sess, ISP15-win, DA60/ID15) handled separately when needed.

Why DiD removes time variation:

- Within-day FE absorbs daily-level confounders (weather, demand, fuel, regulatory shocks).
- Critical and flat hours come from the same dates — calendar mix differences out.
- Pre-IDA crit-flat differential serves as the parallel-trends sanity check.
- *The contrast we identify is mechanism-targeted:* “hours where granularity should bite” vs “hours where it shouldn’t”, not “post-reform vs pre-reform”.

Heterogeneity: firm-level response is not uniform. Naturgy carries most of the result; Iberdrola contributes through nuclear; Endesa and HC are essentially placebo. We see this as supporting the design — only firms with the right portfolio respond.

Bid-shape: a smoking gun that justifies the design

Before showing supply-quantity results, a cleaner piece of evidence that operators *actually use* the granularity strategically: bid-shape behaviour at the unit level, post-MTU15-DA.

What we observe. For Big-4 CCGT in critical hours, the qty-weighted average price is *lower* than in flat hours, but the ladder is *steeper* and has *more tranches*. The firm uses the within-hour granularity to **insure the cheap end (low-price inframarginal tranches lock clearing) and speculate on the expensive tail (steeper ramp captures scarcity rent)**.

Heavy-tailed view: P90 of CV across the 4 quarters of a unit-hour is **4.9×** larger in critical hours than in flat hours — for CCGT.

Tech selectivity. The pattern is concentrated in CCGT and (mildly) hydro — the dispatchable techs where ramping costs make 15-min granularity meaningful. Wind, solar, nuclear show no engagement: P90 ratios near 1×

Why this matters for the design. Bid-shape is unambiguous evidence that operators *can and do* use within-hour granularity. This is the proof-of-mechanism that motivates testing supply-quantity behaviour with a within-hour DiD. The mechanism story is not hypothetical — it is visible in the bid data.

Data pipeline

Coverage: 8 years (2018-01 → 2026-04), at native MTU60/MTU15 granularity throughout.

Three sources, kept strictly separate:

- **OMIE (Iberian market operator):** prices and *bid-level* data for DA, IDA, continuous; per-firm and per-unit programmes. **~5.7 billion rows** (~413 GB raw, ~198 GB processed). Headlined by 3.3 B continuous-market programme rows, 314 M IDA bid tranches, 172 M DA bid tranches.
- **ENTSO-E Transparency Platform:** imbalance volumes and prices, FRR settlement (A87), generation by type/unit, load and renewable forecasts. All system-level fundamentals come from here.
- **ESIOS (REE):** Spanish per-segment settlement, real-time prices, balancing offers. Public-archive endpoints work; **per-BSP/BRP API token still pending**, 3-week wait. Reserves attribution at firm level partly blocked.

Engineering principles:

- Pipelines are **idempotent**: re-running any step produces identical output.
- 25+ data families, each with its own parser; tests; manifest of downloads.
- All paths are repo-relative; data lives on a dedicated external SSD.
- Ledger of ~70 empirical findings tracked formally with *alive/wounded/dead* status and a robustness audit trail.

Headline result: q_2 V-shape

Outcome: q_2 = signed Big-4 IDA repositioning per firm-day-hour, MWh.

Treated: critical hours $h\{7, 8, 16, 17, 18\}$. **Control:** flat hours $h\{3, 4, 5\}$.

Pre: 2018-01 $\rightarrow$ 2024-06-13. **Post:** 2025-10-01 $\rightarrow$ now.

Coefficient	Estimate	SE	p-value
crit_pre (pre-IDA crit–flat gap)	+41.21	3.36	1.6×10^{-34}
DiD δ (crit $\times$ post)	+58.67	12.18	1.5×10^{-6}
crit_post (DA15/ID15 crit–flat gap)	+99.80	—	—

$N = 78,177$ firm-day-hours; cluster-robust SE by date.

Reading. At DA15/ID15, Big-4 firms reposit +100 MWh more in IDA per firm-hour in critical hours than in flat hours. The reform-attributable component (DiD δ) is +59 MWh. Direction matches Allaz–Vila / Ito–Reguant: granularity activates strategic intra-hour repositioning where it has economic content (steep ramps).

Robustness checks (in words)

What I have tested for the +100 headline:

- **Same-calendar-month restriction:** pre-IDA restricted to Oct–May (matches DA15/ID15 calendar). The DiD δ survives at 96% of the full-sample magnitude. Calendar mix is not driving the result.
- **Baseline-window sensitivity:** re-estimated with five different pre-IDA windows (full / excluding 2022–23 crisis / recent 2y / 1y / 6m). The within-DA15/ID15 cross-sectional fact (+100) is *constant across all five windows*. The reform-attribution component (DiD δ) ranges +27 to +133 — pre-trend in crit-flat is real and not stable.
- **Per-hour decomposition:** each hour run individually as treated. Reveals two mechanism clusters with opposite signs — ramp and evening peak hours show positive DiD, midday plateau hours show negative DiD. Granularity acts *bidirectionally*.
- **Cumulative critical-set sensitivity:** top-1, top-2, top-3, top-4, top-5 hours. The +100 result is robust to dropping any single canonical hour. Adding the 6th-ranked hour (*h12*, midday) collapses the coefficient because *h12* belongs to the opposite-sign cluster.
- **Conditional parallel trends:** tested with Spanish wind+solar+demand $\times$ hour-of-day controls. *Does NOT hold*. Renewable controls absorb only $\sim 13\%$ of the pre-trend instability.
- **Bid-shape evidence:** independent confirmation of the granularity mechanism at the unit level (richer bid ladders in critical hours, CCGT only).
- **Extensive-margin:** 13 Big-4 CCGT units inactive in DA60/ID15-prebo critical hours; return under reforzada (REE compulsion via PO-3.x).

Honest assessment

What is robust:

- The within-DA15/ID15 fact: critical hours have **+100 MWh more Big-4 IDA repositioning per firm-hour** than flat hours. Constant across all baseline windows, control sets, conditional specs.
- **Bid-shape rich-ladder pattern** for CCGT post-MTU15-DA. Independent of supply-quantity story.
- **Two-cluster mechanism**: ramp+peak hours positive; midday plateau negative. Granularity is bidirectional.
- **CCGT extensive-margin composition shift** during the asymmetric-clock window.

What is not robust:

- The reform-attributable component (DiD δ) is a *range*, not a point estimate. Pre-trend instability in the crit-flat differential is real.
- **Conditional parallel trends with renewables + demand $\times$ hour does NOT hold**. Some unobserved confounder (CCGT/coal retirement, regulatory anticipation) is also moving the pre-trend.
- Heterogeneity: most of the +100 is carried by Naturgy. Cross-firm response is uneven — a feature for placebo logic, but it complicates the headline.

The theoretical-model gap: the design assumes *in equilibrium*, MTU15 has no economic content for flat hours (otherwise flat is not a valid control). I need a small theoretical model to make this argument explicit. **Out of scope:** a fully estimated structural auction model. **In scope:** a simplified IO model that motivates the empirical design and connects to Allaz–Vila / Ito–Reguant.

What is left

Empirical:

- Build CCGT/coal capacity controls and re-test conditional parallel trends.
- Try regression discontinuity around the sharp Oct 1 2025 boundary.
- Event-study with explicit pre-trend modelling.
- Per-segment settlement once ESIOS API token clears.

Theoretical & writing:

- Simplified IO model justifying the within-day flat-vs-critical design.
- Connect formally to Allaz–Vila and Ito–Reguant.
- Drafting starts after this talk; deadline mid-June 2026.
- **Risk:** tight timeline. Empirical work and theoretical motivation in parallel.

Thank you!

Pablo Paramio Pérez — CEMFI — May 2026

Appendix A1 — Identification: within-day critical-vs-flat hours DiD

The core design idea. The MTU15 reform doesn't change market rules uniformly across hours of the day — it changes them *differentially*. Granularity reform (60-min $\rightarrow$ 15-min) only bites economically in hours where the within-hour residual-demand profile is steep enough that 15-min slicing creates strategic optionality. In flat hours, 15-min and 60-min are mechanically equivalent.

Within-day treatment-control structure:

- **Treated unit:** critical hours $h \in \{7, 8, 16, 17, 18\}$. Top 5 by within-hour residual-demand variance (σ_{within}^2). Steep ramp hours where granularity should bite.
- **Control unit:** flat hours $h \in \{3, 4, 5\}$. Pre-dawn hours, bottom of σ_{within}^2 ranking. Genuinely flat within-hour profile, no strategic optionality from granularity.
- **Treatment timing:** reform regime indicator (DA15/ID15 vs pre-IDA).
- **Estimand:** the causal effect of the reform on the granularity mechanism, identified by how the critical-flat differential changes from pre-IDA to DA15/ID15.

Specification:

$$q_{ift} = \alpha + \beta_1 \text{crit}_h + \beta_2 \text{post}_t + \beta_3 (\text{crit}_h \times \text{post}_t) + \gamma_f + \delta_d + \varepsilon_{ift}$$

Big-4 firm f , day t (Day-of-Week FE δ_d), hour h . Cluster-robust SE by date.

Appendix A2 — Five identification advantages over standard reform DiD

1. **Within-day FE absorbs all daily-level shocks.** Critical and flat hours of the same day share the same Spanish wind regime, the same temperature, the same hydro inflow, the same fuel prices, the same regulatory announcements. Anything operating at the daily level differences out by construction.
2. **Calendar mix controls itself.** Critical and flat hours come from the same dates. Identical calendar mix for treated and control by construction.
3. **Weather differences out at the hour-pair level.** Same-day hours share most weather conditions.
4. **Built-in parallel-trends check.** The pre-IDA crit-flat differential is the placebo test: if pre-existing crit-flat structural differences are large and unstable, the design is in trouble.
5. **Treatment assignment is mechanism-targeted.** The contrast is “hours where granularity should matter” vs “hours where it shouldn’t” — not “post-reform vs pre-reform”. Identifies the granularity-mechanism causal effect specifically.

What within-day FE does NOT solve: within-hour-of-day trends across calendar time. Spanish solar grew $\sim 6\times$ since 2018; renewable share at $h\{7,8,16,17,18\}$ vs $h\{3,4,5\}$ has a calendar-time trend. Conditional parallel trends would require absorbing this. Tonight’s test: it does *not* hold even with $\text{wind} + \text{solar} + \text{demand} \times \text{hour}$ controls.

Appendix A3 — Headline empirical result: q_2 V-shape

Outcome variable: q_2 = signed Big-4 IDA repositioning per firm-day-hour, MWh. From PIBCIE, summing IDA-cleared assigned_power_mw $\times$ mtu_minutes/60 across periods within each (date, firm, hour). Positive q_2 = firm sells more in IDA above DA commitment; negative = firm buys back.

Spec: $q_2 \sim \text{crit} + \text{post} + \text{crit} \times \text{post} + \text{firm FE} + \text{DOW FE}$, cluster by date. Treated $h \in \{7, 8, 16, 17, 18\}$, control $h \in \{3, 4, 5\}$. Pre-IDA = before 2024-06-14. Post = DA15/ID15 (Oct 2025+).

Result with clean flat-hour controls:

Coefficient	Estimate	SE	p-value
crit_pre (pre-IDA crit-flat gap)	+41.21	3.36	1.6×10^{-34}
DiD δ (crit $\times$ post)	+58.67	12.18	1.5×10^{-6}
crit_post (DA15/ID15 crit-flat gap)	+99.80	—	—

Interpretation. At DA15/ID15, Big-4 firms reposit **+100 MWh more** per firm-hour in IDA in critical hours than in flat (pre-dawn) hours. The reform-attributable component (DiD δ) is +59 MWh.

Sample: 78,177 firm-day-hours. Two firms (GN, IB) carry most of the result; see slide 22.

Appendix A4 — Same-calendar-month robustness

The concern this addresses. Pre-IDA spans 2018–2024 (all calendar months). DA15/ID15 spans Oct 2025+ (only Oct–May months). Calendar mix differs $\Rightarrow$ Spanish seasonality could bias the comparison.

The fix. Restrict pre-IDA observations to the same calendar months as DA15/ID15 (Oct–May only). Re-estimate the same DiD spec.

Outcome: q_2 as above (Big-4 IDA repositioning per firm-hour, MWh). Note: this run uses the original definition with all-non-critical-hours as control (the +83 baseline rather than +100 with clean controls).

Pre-IDA window	N_pre	DiD δ	SE	p-value	ratio vs full
Full (default)	224,061	+79.36	6.71	3×10^{-32}	100%
Same-cal-month	—	+76.5	6.77	1×10^{-29}	96%

Verdict. The headline coefficient survives same-calendar-month restriction at 96% magnitude. Calendar mix is not driving the result.

CLAUDE.md verdict criterion (“>50% magnitude with same sign”) easily met.

Appendix A5 — Baseline-window sensitivity (pre-trend stability)

The concern this addresses. The 6.5-year pre-IDA window averages over many distinct conditions (pre-COVID, COVID, energy crisis, normal). If the pre-IDA crit-flat gap drifted, the DiD identification is fragile.

Test. Re-estimate the DiD with progressively narrower pre-IDA windows. With clean flat controls $h\{3, 4, 5\}$:

Pre-window	months	N_pre	crit_pre (SE)	DiD δ (SE)	crit_post
full	78.5	74,700	+41.2 (3.4)	+58.6 (12.2)	+99.8
excl_crisis	54.2	51,414	+72.9 (4.2)	+27.0 (12.4)	+99.9
recent_2y	24.4	23,273	-33.4 (4.4)	+132.6 (12.5)	+99.2
recent_1y	12.2	11,625	-17.1 (5.7)	+116.4 (13.0)	+99.3
recent_6m	6.1	5,778	-13.8 (8.2)	+112.8 (14.3)	+99.0

Two readings:

- **Within-DA15/ID15 fact (crit_post = +99.8) is rock-solid.** Constant across all five baseline choices — this is what the design identifies.
- **Pre-trend is real and not stable.** crit_pre drifts from +73 (deep history) to -33 (recent_2y) — 106 MWh swing. DiD δ ranges +27 to +133 across windows. Reform-attributable component is a range, not a point estimate.

Appendix A6 — Per-hour DiD reveals two-cluster mechanism

Per-hour DiD with control = $h\{3, 4, 5\}$, **outcome** = q_2 . For each hour h , treat that single hour vs the flat-hour control set. Reveals which hours actually carry the reform signal:

Cluster	hours	DiD δ range	p-values	sign
Morning ramp + peak	h7, h8	+64 to +68	$< 10^{-6}$	+
Evening peak	h17, h18, h19	+68 to +101	$< 10^{-7}$	+
Midday plateau	h10–h15	−48 to −131	$< 10^{-7}$	−
Late-night transition	h0–h2	+25 to +46	$< 10^{-4}$	+ (weak)
Other	h6, h9, h16, h20–h22	near zero	ns	—
h23 (late edge)	h23	−70	5×10^{-6}	−

Two-cluster mechanism:

- **Ramp + evening-peak hours: positive DiD δ .** Big-4 reposit MORE in IDA under DA15/ID15. Consistent with Ito–Reguant: residual demand rising sharply within the hour creates strategic withholding amplified by 15-min granularity.
- **Midday plateau (h10–h15): negative DiD δ , large magnitude.** Big-4 reposit LESS (or buy back) in IDA. Solar abundance crowds out conventional commitment; 15-min granularity allows precise quarter-by-quarter reduction in commitment.

Implication. The mechanism is bidirectional — granularity amplifies critical-hour repositioning, in both signs depending on whether residual demand is rising (scarcity) or falling (abundance) within the hour.

Appendix A7 — Cumulative critical-set sensitivity

Why this matters. The canonical 5-hour critical set comes from a top-5 cut of σ_{within}^2 . Why stop at 5? The σ_{within}^2 ranking after rank 5 is $h\{12, 6, 9, 11, 13, 14, 10\}$. Adding any of those should clarify if 5 is the right cut.

Cumulative DiD with control = $h\{3, 4, 5\}$, outcome = q_2 :

Top K critical hours	added	DiD δ (SE)	crit_post	p-value
K=1: {17}	—	+67.7 (16.7)	+113.5	5×10^{-5}
K=4: {7,16,17,18}	ramp + evening	+57.4 (12.7)	+100.5	6×10^{-6}
K=5: canonical {7,8,16,17,18}	+ h8 (morning)	+58.6 (12.2)	+99.8	2×10^{-6}
K=6: + h12	adds midday	+30.5 (11.6)	+71.1	9×10^{-3}
K=11: + $h\{6,9,11-14\}$	full midday + ramp neighbours	-11.4 (10.2)	+26.7	ns
K=12: + h10	also midday	-14.5 (10.1)	+23.5	ns

Verdict. The +100 result is robust to dropping any single canonical hour, and to cutting at K=4 (without h8) or K=1 (just h17). **Adding h12 (rank 6) collapses the coefficient by 30%**, because h12 is in the opposite-sign midday cluster. The σ_{within}^2 ranking is a necessary but not sufficient hour selector — it doesn't distinguish “hours where granularity creates rising-demand scarcity” from “hours where granularity creates falling-demand abundance”. The canonical 5-hour set happens to align cleanly with the rising-demand cluster.

Appendix A8 — Conditional parallel trends test

The test. If the pre-trend in crit_pre (slide 16) is driven by Spanish solar capacity growth, demand growth, or wind variability $\times$ hour-of-day, then conditioning on these should stabilise the DiD δ across baseline windows.

Specs run:

- A: sparse (firm + DOW FE only)
- B: + wind_GWh + solar_GWh + each $\times$ critical (interactions with treated dummy)
- C: + demand_GWh + demand $\times$ critical (additional)

Wind/solar from ENTSO-E A75; demand from ENTSO-E load_actual. UTC $\rightarrow$ Madrid local time, aggregated to hour.

DiD δ across baseline windows, by spec:

Spec	full	excl_crisis	recent_2y	recent_1y	recent_6m	range
A: sparse	+58.7	+27.1	+132.6	+116.4	+112.8	106
B: + wind, solar ($\times$ crit)	+50.4	+28.7	+121.5	+107.2	+107.4	92
C: + ren + demand ($\times$ crit)	+40.1	+10.6	+123.6	+97.2	+101.2	113

Verdict: conditional parallel trends does NOT hold. Renewable controls absorb only $\sim 13\%$ of the pre-trend instability (range 106 $\rightarrow$ 92). Demand controls inflate SEs heavily without further stabilising point estimates. The pre-trend is real and is driven by something my candidate controls don't capture — candidates: CCGT/coal retirement, Rule 28.8 anticipation, structural shifts in dispatchable supply.

Appendix A9 — Per-firm × tech decomposition

Outcome: q_2 apportioned to (firm, tech) via DA-sell-share weighting. crit-flat differential at DA15/ID15 in MWh per firm-hour:

	CCGT	Hydro	Nuclear	Firm total
GN	+163.5	+149.1	—	+312.6
IB	+2.5	+8.6	+84.1	+95.2
GE	−0.7	+5.2	+6.0	+10.5
HC	−24.2	+9.2	—	−15.0

Reading.

- **Naturgy carries the result.** GN's CCGT and Hydro both engage strongly in critical-hour IDA repositioning. GN total +313 dwarfs every other firm.
- **Iberdrola contributes through Nuclear.** IB's CCGT (+2.5) and Hydro (+8.6) are essentially zero. The +95 IB total is +84 from the nuclear-apportioned slice. **Caveat:** this is firm-level q_2 multiplied by IB's DA nuclear sell-share, NOT direct evidence of nuclear-plant-level repositioning. The apportionment is mechanically large for nuclear because nuclear is large in IB's DA portfolio.
- **GE near-zero (+10.5); HC slightly negative (−15).** Functionally placebos: small market share, no IR-amplification response.

Implication for narrative. The granularity-amplifies-IR mechanism appears strongly in one firm (GN) and weakly through one tech in another (IB Nuclear). Cross-firm response is heterogeneous. “Big-4 strategic IDA repositioning” overstates uniform engagement; the data say “Naturgy responds; IB partially through nuclear; GE/HC do not”.

Appendix A10 — CCGT bid-shape: rich-ladder pattern post-MTU15-DA

Independent of the supply-quantity story. Post-MTU15-DA (Oct 2025+), Big-4 CCGT bid LADDERS differ qualitatively across the 4 quarters of an hour — and the difference concentrates in critical hours.

Per-quarter bid measures (CCGT, qty-weighted), critical vs flat:

Measure	flat	critical	direction
Number of price tranches per quarter	5.08	5.50	+8% in critical
Ladder slope $(p_{\text{top}} - p_{\text{bot}})/q_{\text{tot}}$	1.22	1.38	+13% steeper
qty-weighted avg price (€/MWh)	488	472	LOWER in critical

Distributional view. $CV(p_{\text{avg}})$ across 4 quarters of an hour:

Tech	P90 flat	P90 critical	ratio
CCGT	1.6%	7.9%	4.9×
Hydro	1.6%	3.6%	2.3×
Wind	3.1%	3.9%	1.2× (essentially flat)
Solar	0.0%	0.3%	— (very low absolute)
Nuclear	2.4%	2.6%	1.1× (essentially flat)

Interpretation. Pre-MTU15-DA, the firm could only set one price for the hour. Post-MTU15-DA, the firm uses the granularity to bid a richer ladder in critical hours: more low-price inframarginal tranches at the bottom (push avg down, lock clearing) PLUS a steeper high-price tail (capture scarcity rent). **Wind/solar/nuclear do not engage** despite Ito–Reguant downward-curtailment optionality — the strategic story is concentrated in dispatchable thermal/hydro where ramping costs make 15-min granularity economically meaningful.

Appendix A11 — CCGT extensive-margin composition shift (DA60/ID15-prebo)

Number of unique active Big-4 units in critical vs flat hours, by regime (PDBCE auction-cleared):

Regime	flat units	critical units	gap (crit–flat)
pre-IDA	95	95	0
3-sess	72	68	–4
ISP15-win	69	67	–2
DA60/ID15-prebo	53	40	–13
DA60/ID15-reforz	69	66	–3
DA15/ID15	68	66	–2

The 13-unit gap under DA60/ID15-prebo is entirely CCGT. Hydro / nuclear / thermal-nonRE all match across critical and flat (zero or near-zero gap by tech). Decomposed:

Tech (gap crit–flat)	3-sess	ISP15-win	DA60-prebo	DA60-reforz	DA15/ID15
CCGT	–2	–1	–13	–2	–2
Hydro / Nuclear / Other	0	0	0	0–1	0

Reading. 13 Big-4 CCGT units stopped showing up in critical-hour DA auctions during the asymmetric-clock window (Mar 19–Apr 27 2025). Units returned under *operación reforzada* (post-blackout): REE compelled CCGT back into critical hours via PO-3.x voltage-stability requirements. Under symmetric-clock DA15/ID15, units stay in critical hours voluntarily.

Consistent with the asymmetric-clock-induced hedging-cost story: CCGT operators face the worst within-hour mismatch cost (steepest ramps, highest fuel-cost-of-deviation); a portion exit critical hours when the friction is binding.

Appendix A12 — Honest synthesis: what's robust, what's not, what's next

What's robust (identification-clean):

- **Within-DA15/ID15 critical-flat q_2 gap = +100 MWh per firm-hour**, control = $h\{3, 4, 5\}$. Constant across all baseline windows, control sets, conditional specs.
- **Two-cluster mechanism**: ramp + evening peak hours (positive δ); midday plateau (negative δ). Both directions of granularity-mediated repositioning.
- **CCGT bid-shape rich-ladder pattern** post-MTU15-DA. Independent of supply-quantity story; survives on its own. Wind/solar/nuclear do not engage — mechanism is dispatchable-thermal-specific.
- **CCGT extensive-margin composition shift** under DA60/ID15-prebo (13 inactive Big-4 units in critical hours; gap closes under reforzada).

What's not robust (caveats to acknowledge):

- **Reform-attributable component (DiD δ) is a range, not a point estimate**. +27 to +133 across baseline windows. Pre-trend in crit_pre is real (+73 to -33 across pre-IDA sub-windows).
- **Conditional parallel trends with wind+solar+demand $\times$ hour-of-day does NOT hold**. Renewable controls absorb only $\sim 13\%$ of the pre-trend.
- **The +100 is concentrated in Naturgy**. GE/HC are essentially placebo; IB contributes only through nuclear apportionment artifact.

Next steps for the thesis (not for tomorrow):

- Construct CCGT/coal capacity series and re-run conditional spec.
- Regression discontinuity around Oct 1 2025 (sharp MTU15-DA boundary) — avoids the parallel-trends problem.
- Event study with explicit pre-trend modelling.

Disentangle the IB-Nuclear apportionment from physical nuclear repositioning.